Rick Arko

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Full-stack Data Scientist and Machine Learning Engineer specializing in production ML systems, recent focus on forecasting, optimization, and decision automation at the billion-dollar enterprise scale across e-commerce SaaS and 3PL transportation platforms.

TECHNOLOGY

- **Languages:** Python, SQL, Bash, R, TypeScript, Agentic Development (Claude Code, Codex, Cursor, OpenCode)
- **Forecasting & Optimization:** Nixtla, SARIMAX, ETS, Prophet, Probabilistic Forecasting, Hierarchical Reconciliation
- **Machine Learning & Data:** NumPy, SciPy, scikit-learn, PyTorch, AutoGluon, LightGBM, XGBoost, CatBoost, Pandas, Polars, PySpark, DuckDB, MLflow, Optuna, Quantile Regression, Conformal Calibration, MongoDB, SHAP, CrewAI, RAG
- **Platforms:** Databricks, AzureML, Azure, AWS, Docker, FastAPI, Flask, Streamlit, CI/CD, Git, Unity Catalog, Plotly Dash, Vue.js

EXPERIENCE

Arrive Logistics – Chicago, IL

Senior Data Scientist July 2026 - Present

Quantile truckload pricing and agentic tooling for the Data Science org.

- Built quantile pricing models with group and global conformal calibration and automated reporting against production and development
- Built robust experimentation and backtesting pipelines for existing pricing engines and demonstrated improvements with a unified, interpretable alternative
- Stood up Snowflake, Azure ML, FastAPI, Streamlit, and Docker with packaged Claude/Cursor setups for secure tool access across environments
- Created agentic skills that turn scientist notebooks into secure branded apps on shared infra, with no additional cost, and scheduled offline HTML extracts via Slack Bolt

ShipBob – Chicago, IL

Data Scientist / Platform Owner Feb 2022 - May 2026

Technical lead and ML platform owner (Databricks/AzureML) powering forecasting, transit-time prediction, inventory optimization/replenishment, labor planning, and zone-skipping across a billion-dollar e-commerce fulfillment platform delivering hundreds of millions of orders for thousands of customers.

- Delivered a unified ML forecasting engine to replace hundreds of site-specific Prophet/ARIMA models, reducing daily weighted error from approximately 30% to a 15% baseline.
- Built simulation and backtesting engines for forecasting, inventory replenishment, and delivery estimation enabling offline evaluation against strategic KPIs before production release
- Developed a custom delivery estimation solution under tight timelines, to secure onboarding for a multi-billion-dollar account, distilled ML outputs into a static lookup — the solution preserved 80% of modeled lift and demonstrated production scale with Docker Compose and Locust swarms
- Established engineering standards across ~50 scientific repositories (git, CI/CD, linting, testing, backtesting)
- Developed reusable interview case studies used in dozens of senior/staff scientist interviews

Coyote Logistics – Chicago, IL / Remote

Senior Data Scientist / ML Engineer June 2021 - Feb 2022
Data Scientist June 2017 - June 2021
Supply Chain / Data Science Analyst July 2015 - June 2017

Founding member of the Data Science organization, instrumental team builder, individual contributor and tech lead across key strategic initiatives: BookItNow, LoadRec, MarketRate/Cost, Auto Tender Accept, Market-Based Carrier Compensation and more.

- Owned predictive backend for numerous enterprise services with millions of dollars on the line
- Interviewed, hired, and mentored Data Science applicants helping to grow from a department of 2 to over 10
- Developed explainable ML solutions where complexity requires a worthy predictive signal, and distilled SHAP-informed explanations for production outputs (KNN, LightGBM, XGBoost, CatBoost, Random Forest Quantile Regression, etc.)
- Automated statistical forecasting pipelines using SARIMAX and ETS hybrid models
- Automate network optimizations to match empty UPS trucks against Coyote freight

EDUCATION

Lake Forest College - Lake Forest, IL December 2014

Bachelor of Arts in Economics and Mathematics

AWARDS

2nd Place - Software Technology Group AI Hackathon

ShipBob · Dec 2025

Agentic Multi-Modal Demand Forecasting: one-day secure web app for e-commerce cold-start demand prediction with guided agentic feedback, RAG/vector search, and CLIP embeddings. Tools: CrewAI, CLIP, Streamlit, Azure, Nixtla.